Tolga Ensari, PhD

Arkansas Tech University,
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Web: https://www.atu.edu/stem/engineering/cis/profiles/tensari.php
Research Lab: https://ensaritolga.github.io/Portfolio/

Research & Work Experience:

- Assistant Professor at **Arkansas Tech University, AR, USA**, Department of Engineering and Computing Sciences (August 2020, Now)
- Visiting Researcher at **University of Louisville, KY, USA**, Electrical and Computer Engineering, Computational Intelligence Lab (July 2019-August 2020, 1 year)
- Member of the Computational Intelligence Lab (*Director: Prof. Dr. Jacek M. ZURADA*), **University of Louisville**, **KY, USA,** https://engineering.louisville.edu/sites/zurada/computational-intelligence-laboratory/members/
- Assistant Professor at **Istanbul University, Istanbul, Turkey**, Department of Computer Engineering (2009-2020, 10 years)
- Research Assistant at **Istanbul University, Istanbul, Turkey**, Department of Computer Engineering (2001-2009, 8 years)
- Visiting Researcher at **China University of Petroleum**, **Qingdao**, **China,** Computer Engineering, (June, 2018)
- Visiting Researcher at **University of Louisville, KY, USA**, Electrical and Computer Engineering, (July 2014-September 2014, 2 months)
- Visiting Researcher *at* **University of Louisville, KY, USA**, Electrical and Computer Engineering, (September 2011-September 2012, 1 year)
- Visiting Researcher *at* **University of Louisville, KY, USA**, Electrical and Computer Engineering, (June 2011)

Education:

Ph.D. (2003-2009): Istanbul University, Department of Computer Engineering, Istanbul, Turkey

M.Sc. (2000-2003): Istanbul Technical University, Information Technologies, Istanbul, Turkey

B.Sc. (1996-2000): Istanbul Technical University, Department of Electronics and Telecommunication Engineering, Istanbul, Turkey

Research Interests:

- Machine Learning, Deep Learning, Artificial Intelligence (AI)

Research Lab/Group:

- Director of the '*Computational Intelligence and Machine Learning (CIML) Research Lab/Group*' https://ensaritolga.github.io/Portfolio/

Google Research Profile (Citation and H-Index):

- Research profile based on citation, H-index and publications at Google Scholar: https://scholar.google.com/citations?user=o-TS7FsAAAAJ&hl=en&oi=ao

Citations:

Totally **1,602** citations (in Google Research Scholar)

Wikipedia References: "Speckle *Noise", Wikipedia, Wikimedia Foundation, n.d. Web. 17 Nov. 2016*.
https://en.wikipedia.org/wiki/Speckle_noise
https://en.wikipedia.org/wiki/Speckle_(interference)

Funded Research Projects:

1. Project Name : Cancer Detection with Spiking Neural Networks
Funded by : Arkansas Tech University (ATU)
Duration : 6 months (December 2025-May 2026)
Duty : Advisor/Mentor
Budget : $2,000

2. Project Name : Combined Machine Learning and Calphad Approach to Design Multi-Component High Temperature Shape Memory Alloys
Funded by : NASA, EPSCOR Program, Arkansas Space Grant Consortium (Grant Number: RID22015)
Duration : 1 year (July 1, 2022-June 30, 2024)
Duty : CoPI
Budget : $12,048

3. Project Name : Programming Sequence Improvement Program (PSIP)
Funded by : Arkansas Tech University (ATU) Innovation Grant Project
Duration : 2 years (January 2022, January 2024)
Duty : CoPI
Budget : $110,496

4. Project Name : Development of Interdisciplinary Research Collaborative to Provide Datasets in Support of Education Research in Data Science
Funded by : National Science Foundation (NSF)
Duration : 2 years (October 2021-October 2023)
Duty : Researcher
Budget : $99,760

5. Project Name : Summer Undergraduate Research Experience (SURE) Program at Arkansas Tech University
Funded by : National Science Foundation (NSF), Data Analytics That Are Robust and Trusted DART (Award Number: 1946391, # 21-EPS4-0039)
Duration : 3 months (June 1st, 2021-August 31st, 2021)
Duty : CoPI
Budget : $24,000

6. Project Name: Data Analytics that are Robust and Trusted (DART)
Funded by : National Science Foundation (NSF) (NSF Award Number 1946391)
Duration : 5 years (2020-2025)
Duty : Researcher
Budget : $24 million

7. Project Name: Automatic Interface Code Generator from Visual Design Using Machine Learning
Funded by : TUBITAK, Scientific and Technical Research Council of Turkey (Project No: 5200001)
Duration : 2 years (July 2020, July 2022)
Duty : Principal Investigator (PI)
Budget : $46,000

8. Project Name : Smart Digital Marketing and Business Intelligence Development
Funded by : TUBITAK 2244 Program, Scientific and Technical Research Council of Turkey (Project No: 1189B441900333)
Duration : 4 years (August 2020, August 2024)
Duty : Principal Investigator (PI) and Project Director (PD)
Budget : $185,000

9. Project Name : Detection and Classification of Lung Cancer with Machine Learning Methods
Funded by : Council of Higher Education - International Mevlana Project Collaborated with Blida1 University, Blida, Algeria (Project No: MEVLANA-863)
Duration : 1 year (October 2018 - October 2019)
Duty : Principal Investigator (PI)
Budget : $10,000

10. Project Name : Investigation of Mitochondrial DNA Variants with Data Mining in Oxidative Phosphorylation System Diseases
Funded by : Istanbul University, Scientific Research Projects Unit (Project No: TYL-2019-33447)
Duration : 1 year (March 2019 - March 2020)
Duty : Researcher
Budget : $3,000

11. Project Name: Nonnegative Matrix Factorization Applications
Funded by : Istanbul University, Scientific Research Projects Unit (Project No: YADOP-43529)
Duration : 1 year (June 2014 - June 2015)
Duty : Principal Investigator (PI)
Budget : $7,000

12. Project Name : Stability Analysis of Neural Networks with Time Varying Delays and Their Simulations
Funded by : Istanbul University, Scientific Research Projects Unit (Project No: YADOP-5810)
Duration : 1 year (September 2010 - September 2011)
Duty : Principal Investigator (PI)
Budget : $8,500

13. Project Name: New Analytical Methods for Stable Cellular Neural Networks Template Design and Signal and Image Processing Applications of These Templates with Analog CNN-Computer
Funded by : TUBITAK, Scientific and Technical Research Council of Turkey (Project No: 104E024)
Duration : 5 years (April 2004 - April 2009)
Duty : Researcher
Budget : $100,000

Invited Speaker/Research Talks:

- Arkansas Tech University (ATU) Faculty Life Event, Tech Talk, Russellville, AR. Title: Intelligent Finance: Trust Management with Real-World Asset Tokenization and AI, May 8, 2025.

- Solamalai College of Engineering, Department of Electronics and Communication Engineering, India. Title: Neuromorphic Computing, September 15, 2025.

- LISA Academy, Arkansas STEM Festival, Little Rock, AR. Title: Rise of the Machines-Exploring AI's Impact on STEM, March 28, 2024.

- LISA Academy, Rogers/Bentonville Parent Academy, Bentonville, AR. Title: AI Center at LISA Academy, October 30, 2024.

- Arkansas Tech University (ATU), Graduate College, Russellville, AR. Title: Redesigning Higher Education after the AI Revolution, October 21, 2024.

- LISA Academy, Career Day, Fayetteville, AR. Title: AI for the Kids and Education, October 14, 2024.

- LISA Academy, Professional Development (PD) Week, Fire Up'2023, Rogers/Bentonville, AR. Title: AI in Education, August 4, 2023.

- North American University, University Opening Ceremony and Professional Development Activity, Houston, TX. Title: Artificial Intelligence and Ethics, August 14, 2023.

- Arkansas Summer Research Institute (ASRI), NSF/Data Analytics that Robust and Trusted (DART) Project. Title: How to Read a Research Paper, June 8, 2023.

- Arkansas NSF EPSCoR/DART Annual All-Hands Conference, Little Rock, AR. Title: Data Science and Neuromorphic Computing (In-memory Computing/Memristor), May 16-17, 2022.

- The International Conference on Intelligent Systems and Pattern Recognition (ISPR), Hammamet, Tunisia, Title: Machine Learning and Neuromorphic Computing, March 24-26, 2022.

- Arkansas Tech University (ATU), Department of Computer and Information Science. Title: Machine Learning Application Domains, April 4, 2021.

- MagniMind Academy, Data Science Institute, Silicon Valley, Palo Alto, CA. Title: Artificial Intelligence (AI) in Healthcare, July 29, 2021.

- Northern Kentucky University, Department of Computer Science, Highland Heights, KY. Title: Developing New Approaches in Data Science for Healthcare and Applied Mathematics, March 5, 2021.

- Ericsson R&D Department, Istanbul, Turkey. Title: Machine Learning Strategies for Software Development, February 12, 2019.

- China University of Petroleum, Computer Engineering, Qingdao, China. Title: Pattern Recognition and Its Applications on Predictive Churn Analysis, Hyperspectral Image Classification, and Flow Cytometry Data Analysis, June 7, 2018.

- Halmstad University, Department of Computer Science Engineering, Halmstad, Sweden. Title: Data Science, May 15, 2017.

- Istanbul University, College of Medicine, 3D Medical and Industrial Design Lab, Istanbul, Turkey. Title: AI in Healthcare, October 20, 2017.

- Istanbul University, TechnoCity, Istanbul Project Inc., Istanbul, Turkey. Title: How to Write a Research Proposal, August 24, 2017.

- IEEE International Conference on Cloud System and Big Data Engineering, Noida, India. Title: Machine Learning, January 14-15, 2016.

- Freie University of Berlin, Department of Mathematics and Computer Science Engineering, Berlin, Germany. Title: AI for Future, June 11, 2014.

- Indiana University Purdue University-IUPU, Computer and Information Science Department, Indianapolis, USA. Title: Nonnegative Matrix Factorization, August 27, 2012.

- University of Kentucky, Department of Computer Science, Lexington, KY. Title: Pattern Recognition, August 22, 2012.

- Western Kentucky University, Department of Mathematics and Computer Science, Bowling Green, KY. Title: Clustering Algorithms, August 29, 2012.

- University of Louisville, Electrical and Computer Engineering, University of Louisville, Louisville, KY. Title: Research presentation, February 10, 2012.

- University of Louisville, Electrical and Computer Engineering, University of Louisville, Louisville, KY. Title: Research presentation, November 23, 2011.

Scientific Research Projects Consultancy (TUBITAK-TEYDEB Program):

- Albert Solino Software & Consultancy, Istanbul, Turkey
- Veri Software Inc., Izmir, Turkey
- MIRSIS Information Technologies, Istanbul, Turkey
- Cloud Factory Software, Istanbul, Turkey
- MAESTRO Software, Istanbul, Turkey
- MNM Technology, Istanbul, Turkey
- BORAN Software, Istanbul, Turkey
- LUDENS Financial Technologies, Istanbul, Turkey
- HUAWEI Turkey, Istanbul, Turkey
- DGR Information Technologies, Istanbul, Turkey
- OBLOGI Social Media IT, Istanbul, Turkey
- INFOVAS Internet and Telecommunication, İstanbul
- SEMANTIK IT Inc., Istanbul, Turkey
- TURKGEN IT Inc., Istanbul, Turkey

Honours, Awards, and Scholars:

- Nomination and acceptance to the Arkansas Tech University's *Why Tech Wednesday Social Media Program*, September 2023.
- Erasmus+ Scholar, Halmstad University, Computer and Electrical Engineering, Halmstad, Sweden, May 2017.
- Erasmus+ Scholar, Free University of Berlin, Institute for Informatics, Berlin, Germany, May 2014.
- The Council of Higher Education, PostDoc Study Scholar, 1 year (September 2011- September 2012).
- Successful International Researcher Certificate, given by Istanbul University, 2007.
- Successful International Researcher Certificate, given by Istanbul University, 2006.

Scientific Review of Journals:

- Springer Nature
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Automatic Control
- IEEE/ACM Transactions on Computational Biology and Bioinformatics
- Neurocomputing
- Neural Networks
- Cambridge University Press
- Neural Processing Letters
- Physica-A
- Neural Computing and Applications
- Transactions on Computers
- PLOS
- Robotica
- Information Fusion
- Current Bioinformatics
- Computational Intelligence and Neuroscience
- TUBITAK-Turkish Journal of Electrical Engineering & Computer Science

- Computational and Applied Mathematics (AASCIT)
- Istanbul University- Journal of Electrical & Electronics Engineering
- Journal of Applied Computer Science
- International Journal of Neural Systems
- Journal of the Faculty of Engineering and Architecture of Gazi University
- Cumhuriyet Journal of Science
- Turkish Studies
- Journal of Intelligence: Theory and Applications
- Journal of Intelligent Systems
- Journal of Frontiers in Applied Mathematics
- Journal of Service Oriented Computing and Applications
- Journal of Cluster Computing
- Journal of Engineering and Science
- European Journal Science and Technology
- Gazi University Journal of Science

Books:

- *Book Chapter*: Saruhan-Ozdag, F., Yiltas-Kaplan, D., and Ensari, T., Detection of Network Attacks with Artificial Immune System. In D. Burgos, M. Vejar, & F. Pozo (Eds.), Pattern Recognition Applications in Engineering, pp. 41-58, Hershey, PA: IGI Global, DOI:10.4018/978-1-7998-1839-7.ch002, January 2020.
- *Book Chapter*: T. Ensari, M. Gunay, Y. Nalcakan, and E. Yildiz, Overview of Machine Learning Approaches for Wireless Communications, IGI Global International Pub., ISBN: 978-1522574583, USA, January 2019.
- *Reviewer* at De Gruyter (Versita) Book Publishing-Computer Science, Germany, 2014.
- *Writing Lab Manual*: T. Ensari, MATLAB User Guide, 2002 (Link: http://web.itu.edu.tr/kents/matlab.pdf).

Courses Taught:

Machine Learning, Artificial Intelligence, Data Science, Data Mining, Algorithms and Programming, Discrete Mathematics, Database Management Systems, Parallel and Distributed Computing, Compiler Design, Programming Embedded Systems, Organization of Programming Languages, Systems Software and Architecture, Computer Organization and Programming, Circuit Theory, Electronic Circuits, Optimization Techniques, Digital Signal Processing, Microprocessors, Probability Theory and Statistics, Differential Equations, Formal Languages and Automata Theory

Administrative Experience:

- Department/Vice Chairman, Istanbul University, Computer Engineering (September 2014-April 2017)
- Department/Erasmus+ International Exchange Program Coordinator (September 2010 - 2020)
- Department/Mevlana International Exchange Program Coordinator (January 2014 - 2020)

Supervised Master and Ph.D. Thesis:

PhD Thesis (Ongoing):

- Akif Gasi, Thesis title: Content-based Image Retrieval with Machine Learning
  Advisor: Dr. Mustafa Dagtekin, Co-Advisor: Dr. Tolga Ensari

PhD Thesis (Completed):

- Zerrin Yildiz Cavdar, Thesis title: MicroRNA Target Prediction with Machine Learning
  Advisor: Dr. Ahmet Sertbas, Co-Advisor: Dr. Tolga Ensari
- Muberra Akcaman, Thesis title: Microtomography Image Classification with Deep Learning
  Advisor: Dr. Ahmet Sertbas, Co-Advisor: Dr. Tolga Ensari

- Meriem Amrane, Thesis title: Gene Detection from Pathology and Cancer Classification (Prediction) with Machine Learning Techniques
  Advisor: Dr. Saliha Oukid, Dr. Co-Advisor: Dr. Tolga Ensari

- Ebru Aydindag, Thesis title: Cancer Disease Diagnosis with Machine Learning Methods
  Advisor: Dr. Pınar Kirci, Co-Advisor: Dr. Tolga Ensari

- Oznur Sonmez, Thesis title: Cancer Diagnosis from Biomedical Data with Machine Learning Methods,
  Advisor: Dr. Tolga Ensari

Master Thesis (Completed):

- Murat Kazanc, Thesis title: Converting Digital Documents in Image Format to LaTex Code Using Computer Vision and Machine Learning Methods
  Advisor: Dr. Mustafa Dagtekin, Co-Advisor: Dr. Tolga Ensari

- Elif Dogan, Thesis title: MicroRNA Target Prediction for CIRSPR/CAS9 System with Machine Learning
  Advisor: Dr. Ozlem Aktas, Co-Advisor: Dr. Tolga Ensari

- Gurur Pirana, Thesis title: Text Classification with Deep Learning
  Advisor: Dr. Ahmet Sertbas, Co-Advisor: Dr. Tolga Ensari

- Busra Rumeysa Mete, Thesis title: Image Classification with Deep Learning
  Advisor: Tolga Ensari

- Yagiz Nalcakan, Thesis title: Diagnosis of Alzheimer Disease with Deep Learning
  Advisor: Dr. Tolga Ensari

- Eyyup Yildiz, Thesis title: Pattern Recognition in Flow Cytometry Data
  Advisor: Dr. Tolga Ensari. Co-Advisor: Leyla Turker Sener

- Aykut Aydin, Thesis title: Classification of MicroRNA Biogene Sequences with Machine Learning
  Advisor: Dr. Ahmet Sertbas, Co-Advisor: Dr. Tolga Ensari

- Melike Gunay, Thesis title: Predictive Churn Analysis with Machine Learning
  Advisor: Dr. Tolga Ensari

- Fatih Mehmet TEKCAN, Thesis title: Hyperspectral Image Classification with Machine Learning
  Advisor: Dr. Ahmet Sertbas, Co-Advisor: Dr. Tolga Ensari

- Siyabend Turgut, Thesis title: Cancer Diagnosis with Machine Learning Methods
  Advisor: Dr. Mustafa Dagtekin, Co-Advisor: Dr. Tolga Ensari

- Yasin Ugur Belek, Thesis title: Safe Path Planning with Machine Learning
  Advisor: Dr. Tolga Ensari

- Mehmet Salih Kurt, Thesis title: Turkish Question Answering System Using Machine Learning
  Advisor: Dr. Eylem YÜCEL DEMİREL, Co-Advisor: Dr. Tolga Ensari

- İhsan Kocak, Thesis title: Nonnegative Matrix Factorization Applications
  Advisor: Dr. Tolga Ensari

- Feyzan Saruhan Ozdag, Thesis title: Network Intrusion Detection with Machine Learning Techniques
  Advsior: Dr. Derya Yiltas Kaplan, Co-Advisor: Dr. Tolga Ensari

- Rahime Yilmaz, Thesis title: EMG Signal Classification to Predict the Position and Force Patterns of Human Hand
  Advisor: Dr. Tolga Ensari, Co-Advisor: Assoc.Prof. Yunus Ziya Arslan

- Cihan Tiken, Thesis title: Deep Learning Applications
  Advisor: Dr. Tolga Ensari

- M. Metin Tiryaki, Thesis title: Data Mining Based Estimation System for Intelligent Transportation
  Advisor: Dr. Ahmet Sertbas, Co-Advisor: Dr. Tolga Ensari

- Davut Celik, Thesis title: Smart Advertisement/Survey Application Using Data Mining
  Advisor: Dr. Ahmet Sertbas, Co-Advisor: Dr. Tolga Ensari

Membership:

- IEEE Computational Intelligence (CI) Society
- American Association for Science and Technology (AASCIT)
- Hyperdimensional Computing (HD)/Vector Symbolic Architecture (VSA) Society

Editorial Services for Scientific Journals:

- Editor-in-Chief of the Future Digital Technologies and Artificial Intelligence (FDTAI) Journal (Online ISSN: 3070-0965).
- Co-Editor for the Journal of System and Computing (SYCOM).

Scientific Conference Activities:

- General chair of two scientific conferences in AI and machine learning
- Technical program committee (TPC) member of eighty-six scientific conferences in AI and machine learning

Academic/Scientific Publications:

A. Journal Publications:

[1] M. N. Akcaman, T. Ensari, and A. Sertbas, Proposed Mixed Modified Inception and Self Attention U-Net for Medical Image Segmentation: MMIAU-Net, The Computer Science and Engineering Journal, issue 18, no. 1, pp. 19–30, 2025.

[2] A. Gasi, T. Ensari, and M. Dagtekin, Ontology Supported Semantic Based Image Retrieval, Research Square, Journal of Semantic Web, vol. 15, no. 5, pp. 2125-2137, Oct 2024.

[3] Cavdar Z., Sertbas A., and Ensari T., Machine Learning Based Microarray Data Analysis for Target Gene Inference, vol. 17, no. 2, pp. 174-183, 2024.

[4] M. Amrane, S. Oukid, and T. Ensari, Molecules Generation to Inhibit Covid-19 Disease Using Encoder-Decoder LSTM Architecture and PCA Properties, Journal of Mechanics in Medicine and Biology, vol. 23, no. 1, 2023.

[5] M. Kazanc, M. Dagtekin, and T. Ensari, Converting Image Files to LaTeX Format Using Computer Vision, Natural Language Processing, and Machine Learning, ACTA Infologica, vol. 7, no. 2, pp 253-266, October 2023.

[6] Z. Cavdar, T. Ensari, L. T. Sener, A. Sertbas, MicroRNA Target Prediction by Machine Learning-based Microarray Technique: Research Study, European Journal of Science and Technology, Issue 44, pp 39-45, 2022.

[7] T. Ensari, B. Ensari, M. Dagtekin, Violence Detection with Machine Learning: A Sociodemographic Approach, European Journal of Science and Technology, Issue 44, pp. 104-107, 2022.

[8] E. A. Bayrak, P. Kirci, T. Ensari, E. Seven, and M. Dagtekin, Diagnosing Breast Cancer Using Machine Learning Methods, Journal of Intelligent Systems: Theory and Applications, vol. 5, no. 1, pp. 35-41, 2022.

[9] O. S. Sonmez, M. Dagtekin, and T. Ensari, Gene Expression Data Classification Using Genetic Algorithm based Feature Selection, Turkish Journal of Electrical Engineering and Computer Sciences, vol. 29, no. 7, 2021.

[10] A. Gasi, T. Ensari, and M. Dagtekin, Semantic-based Image Retrieval-A Survey, Journal of Acta-Infologica/Istanbul University, vol. 5, no. 2, 2021.

[11] M. Kazanc, T. Ensari, and M. Dagtekin, Filtering Videos by Classification with Deep Learning, IEEE International Congress on Human- Computer Interaction, European Journal of Science and Technology, vol. 26, pp 338-342, Optimization, and Robotics Applications, Turkey, June, 2021.

[12] M. Dağtekin, E. Seven, A. E. Balsever, E. N. Var, L. T. Şener, A. N. Alçalar, B. Ensari, T. Ensari, Post-Traumatic Stress Disorder (PTSD) Management: A Machine Learning Approach, European Journal of Science & Technology, Special Issue, pp. 284-288, August 2020.

[13] T. Ensari, D. C. Armah, A. E. Balsever, M. Dağtekin, Convolutional Neural Networks for Image-Based Digital Plant Phenotyping, European Journal of Science & Technology, Special Issue, pp. 338-342, August 2020.

[14] Ö. Aktaş, E. Doğan, T. Ensari, CRISPR/CAS9 Target Prediction with Deep Learning, Biomedical Journal of Sci&Tech Res, vol. 21, no. 5, Oct 2019.

[15] Aydındağ Bayrak E., Kırcı P., Ensari T., Performance Analysis of Machine Learning Algorithms and Feature Selection Methods on Hepatitis Disease, International Journal of Multidisciplinary Studies and Innovative Technologies, vol. 3, no. 2, pp.135-138, 2019.

[16] [*Invited Paper*] M. Günay, T. Ensari, New Approach for Predictive Churn Analysis in Telecom, Transactions on Communications, vol. 18, pp. 66-70, March 2019.

[17] M. Yankayis, T. Ensari, Performance Evaluation of Feature Extraction Modeling Methods for Speaker Recognition, Journal of Annals of Review and Research, vol. 4, no. 3, pp. 1-11, November 2018.

[18] M. A. Hazar, N. Odabasioglu, T. Ensari, Y. Kavurucu, O. F. Sayan, Performance Analysis and Improvement of Machine Learning Algorithms for Automatic Modulation Recognition Over Rayleigh Fading Channels, Neural Computing and Applications, vol. 29, issue 9, pp. 351-360, May 2018.

[19] T. Ensari, Pattern Recognition from Face Images, International Journal of Naval Science and Engineering, vol. 13, No. 2, pp. 14-20, November 2017.

[20] E. Yıldız, T. Ensari, Gender Classification from Face Images, International Journal of Naval Science and Engineering, vol. 13, No. 1, pp. 31-42, April 2017.

[21] Y. Kavurucu, T. Ensari, Smart Home System Hardware and Software Design, International Journal of Naval Science and Engineering, Vol. 12, No. 2, pp. 40-47, December 2016.

[22] T. Ensari, Character Recognition Analysis with Nonnegative Matrix Factorization, International Journal of Computers, vol. 1, pp. 217-220, September 2016.

[23] T. Ensari, Machine Learning, Information Technologies Magazine, No. 10, pp. 9-10, July 2014.

[24] T. Ensari, S. Arik, New Results for Robust Stability of Dynamical Neural Networks with Discrete time Delays, Expert Systems with Applications: An International Journal 37(8): 5925-5930, 2010.

[25] T. Ensari, S. Arik, Global Robust Exponential Stability of Neural Networks with Multiple Time Varying Delays, Dynamics of Continuous, Discrete and Impulsive Systems, Series A: Mathematical Analysis, Supplementary, Special Issue on Advances in Neural Networks – Theory and Applications, Part I, Vol.14 (S1) pp 6-12, August 2007.

[26] T. Ensari, S. Arik, Global Stability Analysis of Neural Networks with Multiple Time Varying Delays, IEEE Transactions on Automatic Control, Vol. 50, No. 11, pp. 1781-1785, November 2005.

[27] T. Ensari, S. Arik, Global Stability of A Class of Neural Networks With Time Varying Delay, IEEE Transactions on Circuits and Systems - II - Express Briefs, Vol. 52, No: 3, pp. 126-130, March 2005.

B. Conference Publications:

[1] F. Bogrek, H. Sazci, B. Ratermann, M. Guven, R. Shaimurat, B. Uygur, A. Kaya, E. Celebi, E. Erdin, S. Bhandari, and T. Ensari, The AI Center at LISA Academy, Arkansas STEM Posters at the Capitol, Little Rock, January 30, 2025.

[2] S. Bhandari, S. Dhakal, B. Asiroglu, T. Ensari, and B. B. Ustundag, Spiking Neural Networks in Medical Imaging: Revolutionizing Cancer Detection, Arkansas STEM Posters at the Capitol, Little Rock, January 30, 2025.

[3] L. Hong, S. Bhandari, and T. Ensari, Understanding the Impact of Data Poisoning on Financial Fraud Detection: Thresholds and Feature Sensitivity, ATU Student Research Symposium, Russellville, April 29, 2025.

[4] R. Shaimurat, F. Bogrek, B. Ratermann, S. Bhandari, T. Ensari, and J. Wood, Empowering Students through Project-Based Learning in Computer Science: Connecting Real-World Problems to Classroom Innovation, ATU Student Research Symposium, Russellville, April 29, 2025.

[5] S. Bhandari, V. Gogineni, M. Kelestemur, and T. Ensari, AI-Driven Computer Vision: Foundations, Innovations, and Applications, The Society of North American Scholars (SNAS) Interdisciplinary Research Conference, Fairleigh Dickinson University, New Jersey, October 10, 2025.

[6] S. Bhandari, B. Asiroglu, S. Dhakal, R. Ghosh, T. Ensari, and B. b. Ustundag, Spiking Neural Networks for ECG Classification and Anomaly Detection, IEEE 16th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), October 22-24, 2025, NY, USA.

[7] S. Bhandari, S. Dhakal, T. Ensari, B. B. Ustundag, Spiking Neural Networks in Medical Imaging: Revolutionizing Cancer Detection, Arkansas STEM Posters at the Capitol, Little Rock, AR, Jan 30, 2025.

[8] F. Bogrek, H. Sazci, B. Ratermann, M. Guven, R. Shaimurat, B. Uygur, A. Kaya, E. Celebi, E. Erdin, S. Bhandari, T. Ensari, The AI Center at LISA Academy, Arkansas STEM Posters at the Capitol, Little Rock, AR, Jan 30, 2025.

[9] H. Raji, M. Kelestemur, T. Ensari, and S. Saedi, High Entropy NiTiHfZrCu SMA; Transformation Temperatures, Microstructure and Mechanical Performance, TMS 2024, 153rd Annual Meeting & Exhibition, Orlando-Florida, March 3-7, 2024.

[10] S. Bhandari. V. Gogineni, J. Wood, T. Ensari, and B. Ustundag, Anomaly Detection with Spiking Neural Networks, Arkansas Tech University (ATU) Research Symposium, Russellville, April 18, 2024.

[11] M. Corbett-Strain, T. Ensari, B. B. Ustundag, Neuromorphic Computing for Data Compression, STEM at the Capitol, Little Rock, Feb 14, 2024.

[12] R. Tarannum, T. Ensari, Occluded Traffic Sign Detection with YOLOV3, STEM at the Capitol, Little Rock, Feb 14, 2024.

[13] R. Tarannum, J. Wood, and T. Ensari, Breast Cancer Classification with Machine Learning, Arkansas Tech University (ATU) Research Symposium, Russellville, April 18, 2024.

[14] M. Rahman, M. H. Kelestemur, S. Saedi, and T. Ensari, Classification and Segmentation of Material Microstructures Images Using Machine Learning, STEM at the Capitol, Little Rock, February 14, 2024.

[15] A. Zhamanov, T. Ensari, F. Taban, A. Kucuk, and A. Boltayev, AI Task Force Initiative at North American University: Pioneering AI-Driven Transformation in Education, Research, and Administration, Interdisciplinary Research Conference, Society of North American Scholars, Western Connecticut State University, Danbury, CT, October 11, 2024.

[16] Z. Orhan, B. Ensari, T. Ensari, and F. Arslan, Advancing Mental Health Care Through AI and Interdisciplinary Efforts in Higher Education, Interdisciplinary Research Conference, Society of North American Scholars, Western Connecticut State University, Danbury, CT, October 11, 2024.

[17] B. Ensari, J. Wood, and T. Ensari, Enhancing Student Development with AI: A Case Study of Code Monkey in Social Emotional Learning (SEL), Interdisciplinary Research Conference, Society of North American Scholars, Western Connecticut State University, Danbury, CT, October 11, 2024.

[18] M. Rahaman, R. Ghosh, I. Dutta, T. Ensari, and R. Cunningham, Enhancing the Programming Sequence for Undergraduate Computer Science Students: A Program for Improving Learning Outcomes, International Congress on Human-Computer Interaction, Optimization, and Robotic Applications, Istanbul, Turkey, June 8-10, 2023.

[19] M. Rahaman, R. Ghosh, R. Cunningham, I. Dutta, and T. Ensari, Programming Sequence Improvement Program (PSIP), STEM at the Capitol Workshop, Little Rock, February 8, 2023.

[20] M. M. Alam, B. B. Ustundag, D. T. Altilar, and T. Ensari, Classification Analysis with Spiking Neural Networks, Arkansas Tech University Research Symposium, Russellville, USA, April 25, 2023.

[21] A. Zencirli, H. Cetin, N. Tug, and T. Ensari, Improved Next Item Recommendation for Long Tail Products with Machine Learning, International Congress on Human-Computer Interaction, Optimization and Robotic Applications, Ankara, Turkey, June 9-11, 2022.

[22] B. Asiroglu, S. Senan, P. Gorgel, M. E. Isenkul, T. Ensari, A. Sezen, M. Dagtekin, A Deep Learning-based Object Detection System for User Interface Code Generation, International Congress on Human-Computer Interaction, Optimization and Robotic Applications, Ankara, Turkey, June 9-11, 2022.

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